EGS3012S 2025 Introduction to physical Atmospheric Science

How do we study atmospheric variability?

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EGS Building Rm 4.14

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Before We Start...

- What do you expect from this section of the course?
- What I'll be doing
- What can you do if you have concerns?
  1. Discuss with me
  2. Class rep
  3. Dr. Sabina Abba Omar (CSAG)

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1. Week 1: Climate Variability & Change - PM
2. Weeks 2–3: Climate change impacts - PM
3. Week 4–6: Atmospheric thermodynamics
4. Week 7–9: Atmospheric dynamics & circulation
5. Week 10: Weather systems
6. Week 11: SA Weather & forecasting - LvS
7. Weeks 12–13: Atmospheric modelling (SAO) & predictability

Part I

General concepts

Roadmap

What are we doing here?

- What is Atmospheric Science?
- Why study the atmosphere?
- How to study the atmosphere?

What are we doing here?

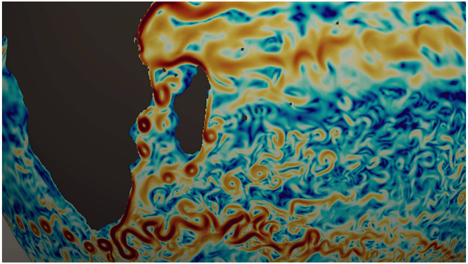
- What is Atmospheric Science?
- Why study the atmosphere?
- How to study the atmosphere? $\Leftarrow$ today's focus

Roadmap

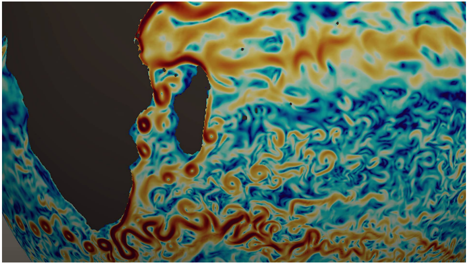
1. Taking a step back
2. Models!
3. Weather & Climate

Modelling in natural sciences

"Truth ... is much too complicated to allow anything but approximations" - John von Neumann



Modelling in natural sciences



"The [physical] sciences do not try to explain, they hardly even try to interpret, they mainly make models. By a model is meant a mathematical construct which, with the addition of certain verbal interpretations, describes observed phenomena. The justification of such a mathematical construct is solely and precisely that it is expected to work."- John von Neumann

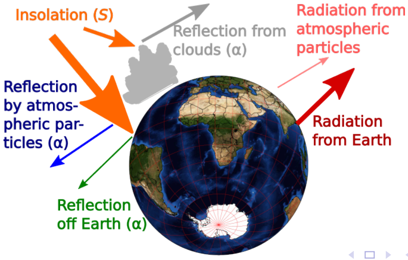
What is a model?

No, not on a catwalk...

- Modelling Earth's atmosphere can be thought of as our primary aim
- Dr. Phikolomzi Matikinca has introduced you to climate change projections, obtained from climate models
- You will hear towards the end of the course form Dr. Sabina Abba Omar about operational weather & climate models, used to forecast tomorrow's weather or project climate change patterns a 100 years ahead.
- Fundamentally, operational models are dynamical models with statistical/ empirical components to parametrise key processes that can't be resolved explicitly
- Understanding and describing the full range of atmospheric (let alone climate system) behaviours together is not feasible...

Building simple models

- We artificially but carefully isolate sections of or processes in the atmosphere that we can meaningfully regard as essentially separate; then build a conceptual model
- Use core physics principles, observations & maths
- Describe the process quantitatively, with the aim of understanding & predicting
- This is an idealised representation of reality
- Assumptions need to be made, with scope of applicability
- Whether a model is good (enough) depends on the question



Key principle

Principle

There's no best model of anything interesting, but there's always a better one

- Any model of reality is incomplete, so will not be perfect and this imperfection has consequences
- Models can be improved by identifying the most severe shortcomings and addressing them
- Thus, we can develop ever more complete, accurate & sophisticated models iteratively.

On Monday, we'll recap some principles and then get into our first model!

Roadmap

The Atmosphere, Weather and Climate

Definition (Atmosphere)

A gaseous layer/envelope around a planet, bound to it by gravity.

Definition (Weather)

The state of Earth's atmosphere at a particular place & time.
Includes temperature, wind, rainfall, clouds, etc..

Definition (Climate)

Statistical description of normal (or average), likely and possible weather over a longer period of time (months to millennia).
Includes variations spatially & by season & time of day.
Alternatively, the state of slowly evolving components of the climate system, evolving at larger spatial scales.

A Systems View: Conceptual model of Earth's climate

(Internal) Components:

Atmosphere	Our focus
Hydrosphere	Includes oceans, lakes and rivers
Cryosphere	Includes sea ice, land ice and glaciers
Land surface	Topography, soil moisture, surface roughness, etc.
Biosphere	All living things

And all their interactions...

A Systems View: Conceptual model of Earth's climate

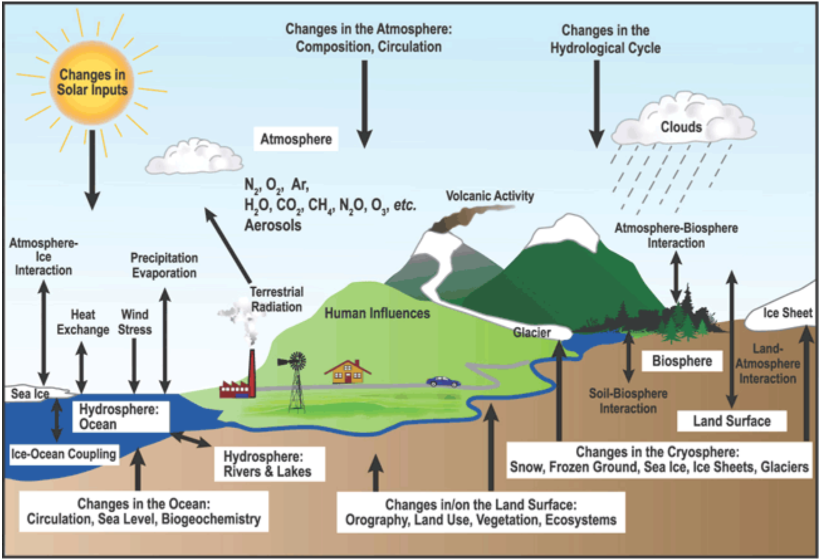


Image credit: IPCC AR4 SPM; note: Human GHG emissions, Solar & volcanic are considered external forcings to climate system

What is "normal"? (relax, this isn't Geographic Thought!)

Saying: "Climate is what you expect, weather is what you get."

- Generally, we expect what is "normal" or "average".
- We also expect variability, but not extreme deviations
- Some months, years are wetter than others
- droughts & floods, though, tend to happen when it's much drier/wetter than average over an extended period.
- When is a deviation extreme? Relevant for your project later...
- Deviations or differences from long-term averages are often called anomalies in climatology
- Anomalies are key for understanding climate variability, change and extremes

Weather vs climate view

Weather: It was cold and wet on Saturday morning on UCT Upper Campus

Climate: The winter of 2023 over the south-western Cape was one of coldest and wettest in many decades

In between? It's been cold & but not very wet this August in Cape Town

- Notice the difference in time- and space- scales.
- Climate $\Rightarrow$ summing, averaging, integrating, finding extremes over time.
- Also, note the reference to times of day & season. What we expect varies with these factors, as well as between different regions and locations on Earth. Why?